# From Inbox to Board: Email Intake Automation

Sep 30, 2026 · @Omar

## Summary

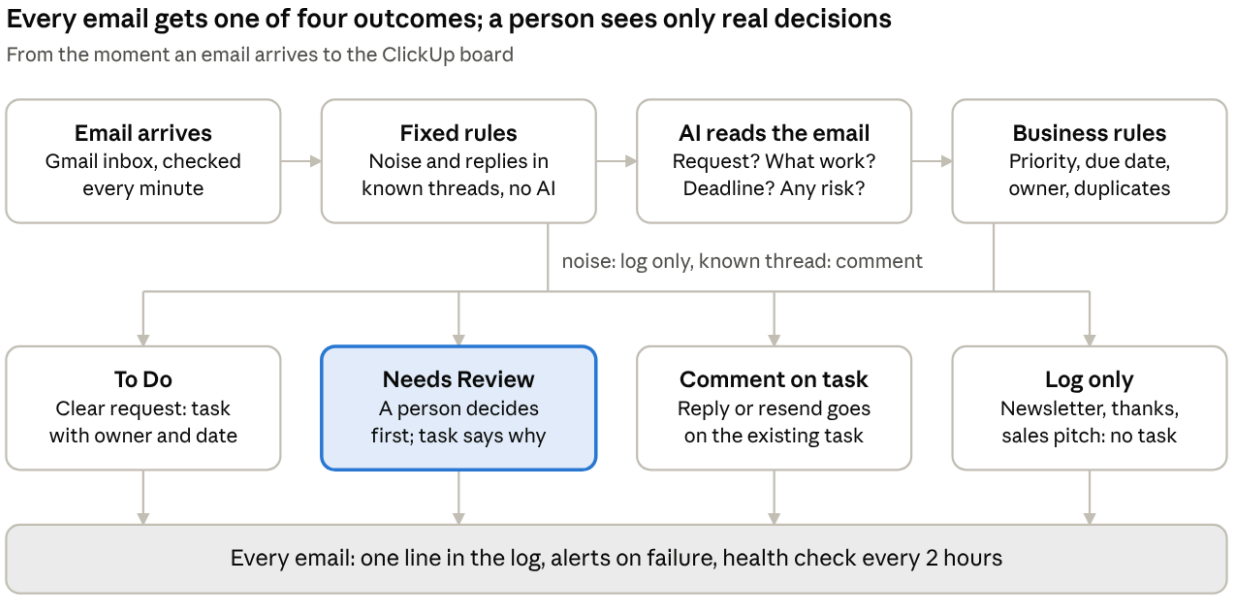
I built a working automation in n8n that turns work emails into ClickUp tasks. It checks the requests inbox every minute and sets each task's title, suggested owner, priority and due date. It targets the three problems in the brief: requests that are missed, logged twice, or given the wrong urgency.

Not every email should become a task automatically. Clear requests go straight to To Do, replies become comments on the existing task, and newsletters, thank-you notes and sales pitches are only logged. A person looks first only when a decision is needed before work starts, and the task says what to decide.

On 20 new test emails, all 20 got the right outcome, and every checked priority, category, owner and due date was correct. Each email takes about 7 seconds to process.

The three n8n workflows are attached as importable JSON files, with a setup guide and the 30 labeled test emails.

## How it works




intake flow · 4 steps, 4 outcomes

Fixed rules handle the obvious cases without an AI call. The AI only reads and proposes; fixed rules make the final call on priority, dates, owner and outcome, so each decision can be tested.

## Should every email become a task?

No. Every email gets exactly one of four outcomes, and a person steps in only when a decision is needed before work can start.

| Outcome | When | What happens in ClickUp |
|---|---|---|
| To Do | A clear work request | A task in the Requests list with title, owner, priority, due date and a summary of the email |
| Needs Review | A person must decide something first (reasons below) | A task in a separate Needs Review list that says why it is there and what to decide |
| Comment | A reply in a known thread, or the same person re-sending the same request | A comment on the existing task, no new task |
| Log only | Newsletters, auto-replies, calendar invites, thank-you notes, FYIs, vendor sales pitches | No task; the email is recorded in the log |

**When a person looks first.** An email goes to Needs Review only for one of these reasons:

1. It is unclear what is being asked, or whether any work is needed at all.
2. The deadline in the email is already in the past.
3. It asks us to approve, pay, sign or commit money or a contract, or it is a legal matter.
4. It is a personal or HR matter about a named person, such as salary or performance.
5. It may repeat a task that already exists.
6. It asks different teams for different things, so it must be split first.
7. The AI step failed, or no owner is set for that type of work.

**Why I drew the line here.** A wrong automatic task costs more than a short review: someone works on the wrong thing, money is committed without approval, or a confidential matter spreads. Reviewing every email would bring the morning bottleneck back. So routine cases are automatic, and only decisions reach a person.

The rule leans towards safety. An email is logged without a task only when the AI is at least 85% sure it needs no work; below that, a person decides.

**The content decides, never the sender.** A clear request from a new client writing from Gmail becomes a task like any other. An unclear email from the General Manager still goes to review. The sender only adds context: the client's name on the task, and at least High priority for the General Manager.

## Urgency and due dates

The AI proposes a priority and a deadline, then fixed rules check both, so urgency follows the same rules every time. Each task quotes the words from the email that set its priority and date.

**Priority**

- **Urgent** only when something client-facing is broken right now, or the work is due today. The words "urgent", "ASAP" or "عاجل" alone never make a task Urgent.
- **High** when the deadline is within 3 business days, for a client escalation, or for a General Manager request.
- **Low** when the sender says there is no rush, for example "no rush" or "مش مستعجلين".
- **Normal** for everything else.

**Due dates**

- Dates are read in English and Arabic, including relative ones. "Tomorrow" or "بكرا" means the next business day, and "end of week" means Thursday.
- The work week is Sunday to Thursday. A written deadline that falls on a weekend moves to the business day before it; a relative one moves to the next business day.
- With no date in the email, the deadline follows the priority: Urgent the same day, High +2, Normal +5 and Low +10 business days. The task says the date was inferred.

## Duplicates and follow-ups

The same request does not become two tasks.

- **A reply in a thread that already has a task** becomes a comment on that task. This is a fixed rule with no AI call.
- **The same person re-sending the same request in a new email**, even reworded or in another language, becomes a comment too when the match is at least 90%. The AI compares the email with the 25 most recent tasks, and a text-similarity check compares the titles.
- **A likely match from a different person**, such as a colleague forwarding a client's email, goes to Needs Review as a possible duplicate, with a link to the existing task.
- **The same email delivered twice** is processed once; the extra copy is only logged.

## Reliability: nothing is lost silently

Every email gets exactly one line in a log with its outcome and the reason for it, so any decision can be traced back.

- **If the AI fails**, the email goes to Needs Review instead of being dropped.
- **If ClickUp rejects an action**, the failure is logged and an alert is raised with the email's details.
- **Alerts:** every failure becomes one alert. It is saved, shown on a monitoring dashboard and emailed when critical, at most once every 15 minutes for the same problem. Each alert says what to do, for example "reconnect the ClickUp login".
- **Health check every 2 hours:** it compares the inbox with the log and raises an alert for any email waiting more than 15 minutes. It also checks that the inbox can still be read. This catches silent failures, such as the workflow being switched off.
- **No loops:** auto-replies and the automation's own emails are ignored.

## Testing and results

I tested against labeled emails with a known correct answer, and the workflow scores itself after every run.

I wrote 48 realistic emails in English, Arabic and mixed language, each labeled with the expected outcome, priority, category, owner and due date. I used 38 for building and held 10 back as a check. Tests can run in a mock mode that writes to a table instead of ClickUp, and each run records the prompt version so results can be compared.

**What the tests changed**

1. **The sender rule.** An early version sent every unknown sender to Needs Review. The tests showed a new client writing from a personal email address held back only because of the address. I replaced the rule: the content decides.
2. **Sales pitches.** A vendor asking to book a 15-minute call became a task to schedule the call. I taught the AI to tell a vendor selling its own service from a real request, and the retest logged both pitches correctly.

**Results on 20 new emails.** After the logic was final, I wrote 20 new emails and ran them in two batches against the live ClickUp workspace. Priority, category, owner and due date are checked only where an email has an expected value; a comment, for example, has no priority.

| Check | Correct |
|---|---|
| Outcome (To Do, Needs Review, comment, log only) | 20 of 20 |
| Priority | 8 of 8 |
| Category | 10 of 10 |
| Owner | 10 of 10 |
| Due date | 8 of 8 |

**Speed and cost.** A batch of 10 emails took about 70 seconds, so about 7 seconds per email. Emails caught by the fixed rules skip the AI entirely. The AI step uses gpt-5-mini, listed at $0.25 per million input tokens and $2 per million output tokens (30 Sep 2026). Assuming 4,000 input and 1,000 output tokens per email, that is about $0.003 per email. At 100 emails a working day, it comes to about $7 a month.

## The alternative I considered: a custom AI agent in code

I considered building the intake as a Python app with LangGraph, where an agent reads each email and calls tools (Gmail, ClickUp, a database) itself. I decided against it for this problem:

- **Slower to deliver:** every piece (Gmail polling, the ClickUp API and authentication, retries, a store for the log and thread links, scheduling, hosting) would have to be written, tested and reviewed from scratch. n8n has them ready.
- **More parts to maintain:** several separate tools wired together, each one something that can break.
- **Less visible:** n8n logs every run step by step, so anyone can open an email's run and see why it went where it went.
- **Harder for the team to change:** in n8n, owners, thresholds and rules change without a developer or a deployment.
- **Slower to extend:** in n8n, adding a feature or changing the scope takes a few nodes and can be tested the same day against the labeled emails. Examples: a new request source like a web form, a Slack alert for urgent tasks, or a new review rule. In code, every change is a full cycle of writing, testing, reviewing and deploying.
- **Not needed here:** this is a fixed pipeline, not open-ended reasoning. The agent flexibility LangGraph offers would add complexity without adding value.

**When I would revisit it:** if the process grows into multi-step conversations, for example asking the requester a clarifying question and waiting for the reply before creating the task.

## Limitations and next steps

These are the known gaps, in the order I would address them.

1. **Failed ClickUp actions are not retried.** They are logged and alerted, and a person creates the task by hand. Next: an automatic retry queue.
2. **Overflow is detected, not reprocessed.** One run handles up to 16 emails, and the health check flags any that were left waiting. Next: feed those emails back into the intake automatically.
3. **A new request inside an old thread becomes a comment.** Replies skip the AI by design. Next: let the AI check replies for a separate new ask.
4. **The AI time limit is tight.** In live Gmail tests, one of five emails hit the 15-second limit and went to Needs Review, as designed. Next: raise the limit to 30 seconds.
5. **The test set is small.** 20 new emails is a small sample. Next: run a week of real inbox traffic in mock mode and measure before switching it on.
6. **Owners are roles, not people.** Tasks show a suggested owner by role. Next: map each role to a ClickUp user so tasks are assigned automatically.
7. **The model is one setting.** gpt-5-mini is marked as legacy on the pricing page cited above, so a newer one can be swapped in after re-running the labeled test set.
8. **The email provider.** The demo reads a Gmail inbox. If the company uses Microsoft 365, the Gmail steps are swapped for Outlook ones and the triage logic stays the same.

## Appendix: the 20 new test emails

All 20 were handled as expected. T01 to T10 are dated Wednesday 30 September 2026 and T11 to T20 Thursday 1 October, which is what the due dates count from.

| # | Email | Result |
|---|---|---|
| T01 | Client asks for a Q3 sales report by Tuesday 13 October | To Do, Normal, due 13 Oct |
| T02 | Client, in Arabic, wants a sale poster "tomorrow" | To Do, High, due 1 Oct |
| T03 | Client's website tracking page is broken, "fix it today" | To Do, Urgent, due 30 Sep |
| T04 | Client replies to T01 asking for one more page | Comment on T01's task |
| T05 | Colleague: "can someone look into the Al-Waha thing?" | Needs Review: unclear ask |
| T06 | Colleague asks us to approve and pay a laptop deposit | Needs Review: money |
| T07 | Colleague asks for a dashboard and a brochure for one client | Needs Review: two teams |
| T08 | Newsletter | Log only, fixed rule |
| T09 | Client, in Arabic, says thank you | Log only |

| # | Email | Result |
|---|---|---|
| T10 | New client from a personal address wants a website quote, "no rush" | To Do, Low, due 14 Oct |
| T11 | Client chases T03 in the same thread | Comment on T03's task |
| T12 | Client, in Arabic, asks about T02's poster in a new email | Comment on T02's task |
| T13 | General Manager asks for a projects summary by 15 October | To Do, High (General Manager), due 15 Oct |
| T14 | Client, in Arabic, wants a brochure change "tomorrow", sent on a Thursday | To Do, High, due Sunday 4 Oct |
| T15 | Client asks us to sign a non-disclosure agreement | Needs Review: legal |
| T16 | Client asks for a report "by 28 September" in an email sent 1 October | Needs Review: deadline in the past |
| T17 | Colleague's out-of-office auto-reply | Log only, fixed rule |
| T18 | Vendor pitches its chatbot service | Log only |
| T19 | Colleague asks to book a room and order lunch for a workshop | To Do, High, due 5 Oct |
| T20 | Client, in Arabic, asks to change a weekly report "next week" | To Do, Normal, due Thursday 8 Oct |